

Theorem and notation sheet (Supplement S1)

[AUTHOR — to be completed by the author]

Working draft

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S1. Theorem and notation sheet

This supplement states the main result and every lemma it uses, defines each symbol exactly once, and gives complete proofs of Theorem 1 and of its lemmas, written to be worked through line by line; Proposition 1 is proved in Supplement S4, and the class-hierarchy propositions in Supplement S3. Every load-bearing definition, equation and statement carries an entry in the build manifest that traces it to the proof record (file, commit, section). Throughout this supplement m_2 is kept symbolic; its numerical value appears only in Section 6 and Supplement S5.

How to read the proof. The argument has one idea and four supporting facts. The idea: a shared signed-affine modulation of one exogenous process cannot change the *absolute* standardised skewness of the force, so if one protocol has zero skewness and another has non-zero skewness, no such shared representation exists. The four facts are: (i) an exact identity that turns cumulants of the bath force into cumulants of a single oscillator (Lemma 1); (ii) a symmetry that makes the reference protocol’s skewness vanish exactly (Lemma 2); (iii) a first-order expansion of the ramp protocol’s third cumulant in $1/N_B$ with a controlled remainder (Lemma 3); and (iv) a small-time computation showing that the first-order coefficient is strictly negative (Lemma 4). Lemma 5 makes sure the standardisation is legitimate, and Lemma 6 is the precise form of the idea.

S1.1 Notation

Symbol	Meaning
$\mathcal{D}$	the model of Definition 1 (finite Duffing bath, reciprocal coupling)
t, T	time; the horizon is $[0, T]$ with $T = 2\pi$
M, p_q, V	mass, momentum and potential of the system coordinate; M denotes the mass only in (S1.1), and the location functional $M_t[q]$ everywhere else
H, H_{int}	parent Hamiltonian (S1.1) and its coupling term, $H_{\text{int}} = -N_B^{-1/2} q \sum_j x_j$
$q(t)$	prescribed (clamped) trajectory of the system coordinate; a <i>protocol</i>
$\mathcal{X}$	class of admissible clamps (Definition 2)
$q_{\text{ref}}, q_{\text{ramp}}$	reference protocol and ramp protocol (Definition 2)

Symbol	Meaning
$s(u)$	smooth ramp polynomial used by q_{ramp}
N_B	number of bath oscillators (bath size)
ε	coupling per oscillator, $\varepsilon = N_B^{-1/2}$
x_j, p_j	position and momentum of bath oscillator $j \in \{1, \dots, N_B\}$
$H_0(x, p)$	single-oscillator energy, (S1.4)
β	inverse temperature; $\beta = 1$ throughout
ρ	single-oscillator Gibbs law at $\beta = 1$, (S1.5)
$z_0 = (a, b)$	initial position and momentum of one oscillator, drawn from ρ
E_0	initial energy $H_0(z_0)$
$X_q^\varepsilon(t)$	position at time t of one oscillator driven by εq
$x_0(t)$	unforced trajectory, $X_q^0(t)$ (it does not depend on q); not an oscillator label
$y_1(t)$	first-order response $\partial_\varepsilon X_q^\varepsilon(t) _{\varepsilon=0}$ for the ramp protocol
$F_q(t)$	interventional environment force under protocol q , (S1.3)
$\hat{F}_q(t), Z_q(t)$	centred and standardised force (Definition 3)
m_k	Gibbs moment $\mathbb{E}[a^k]$ of the initial position
$\text{Var}(x_0^2)$	$m_4 - m_2^2$, the variance of a^2 (equal to that of $x_0(t)^2$ for every t)
κ_n	joint cumulant of order n ; κ_2 is the (co)variance
$\gamma_1[Y]$	standardised skewness $\kappa_3(Y)/\text{Var}(Y)^{3/2}$
$c(t)$	$\text{Cov}(x_0(t)^2, y_1(t))$, (S1.8)
$K(t)$	$3c(t)$, the first-order coefficient of the third cumulant
c_k	Taylor coefficient of t^k in $c(t)$
α_k, η_k	Taylor coefficients of t^k in $x_0(t)$ and $y_1(t)$
Λ	constant in the order-eight Taylor remainder of $c(t)$
$\Theta(t)$	constant in the order-three remainder of the third cumulant in ε
Q, c_Q	bound for a clamp, $ q \leq Q$, and the energy constant $c_Q = \sqrt{2} Q \pi$ (Lemma 3)
$R_N(t)$	remainder in the first-order expansion (S1.11)
$D_8(z_0)$	bound on the eighth time derivative in the proof of Lemma 4
$\delta, N_0(t)$	existential time window and bath-size threshold of Theorem 1
$\mathcal{H}$	harmonic baths with coupling linear in bath and system coordinates (Definition 4)
$\mathcal{E}_1, \mathcal{E}_2^\pm, \mathcal{E}_{\text{univ}}$	competitor classes (Definition 4)
$M_t[q], G_t[q], \xi$	deterministic location, signed scale, shared exogenous process
$s_q(t)$	sign assigned to protocol q at time t by the shared sign functional S_t
$U, \mathfrak{F}$	exogenous random object and causal functional of $\mathcal{E}_{\text{univ}}$ (Definition 4)
Y, B, C, Ψ	state, drift, force map and initial-state map of a deterministic causal environment (Definition 4)
p_0	initial momentum coordinate, $p_0 = b$; also written $\dot{x}_0(0)$
S_t	shared causal sign functional (Definition 4)
D_q	degeneracy set $\{t : \text{Var } F_q(t) = 0\}$
$C_0(t_a, t_b)$	reservoir-limit covariance $\langle x_0(t_a) x_0(t_b) \rangle_{\text{Gibbs}}$, (S1.13)

S1.2 Definitions

Definition 1 (model $\mathcal{D}$). A system coordinate q with momentum p_q and mass M is coupled reciprocally to N_B identical Duffing oscillators,

$$H = \frac{p_q^2}{2M} + V(q) + \sum_{j=1}^{N_B} \left[\frac{p_j^2}{2} + \frac{x_j^2}{2} + \frac{x_j^4}{4} \right] - \frac{1}{\sqrt{N_B}} q \sum_{j=1}^{N_B} x_j, \quad V(q) = \frac{q^2}{2} + \frac{q^4}{4}. \quad (\text{S1.1})$$

The system potential V only defines the autonomous parent model; it plays no role once q is clamped. Under a clamp the oscillators obey

$$\ddot{x}_j + x_j + x_j^3 = \varepsilon q(t), \quad \varepsilon = N_B^{-1/2}, \quad (\text{S1.2})$$

and the environment force on the clamped coordinate is

$$F_q(t) = -\frac{\partial H_{\text{int}}}{\partial q} = \frac{1}{\sqrt{N_B}} \sum_{j=1}^{N_B} x_j(t). \quad (\text{S1.3})$$

The single coupling term produces both the force on the bath ($+\varepsilon q$ per oscillator) and the force on the system, so the coupling is reciprocal. With

$$H_0(x, p) = \frac{p^2}{2} + \frac{x^2}{2} + \frac{x^4}{4}, \quad (\text{S1.4})$$

the preparation is the same for every protocol: at $q(0) = 0$ the coupling vanishes, and the pairs $(x_j(0), p_j(0))$ are independent and identically distributed with law

$$\rho(x, p) \propto e^{-\beta H_0(x, p)}, \quad \beta = 1. \quad (\text{S1.5})$$

Definition 2 (clamps and protocols). The admissible clamps $\mathcal{X}$ are the prescribed trajectories $q : [0, T] \rightarrow \mathbb{R}$ that are C^2 , satisfy $q(0) = \dot{q}(0) = 0$, have bounded q , $\dot{q}$, $\ddot{q}$, and are applied to the one fixed preparation (S1.5). Two members of $\mathcal{X}$ are used:

$$q_{\text{ref}}(t) \equiv 0, \quad q_{\text{ramp}}(t) = \begin{cases} s(t/\pi), & 0 \leq t \leq \pi, \\ 1, & \pi < t \leq 2\pi, \end{cases} \quad s(u) = 10u^3 - 15u^4 + 6u^5. \quad (\text{S1.6})$$

Only the restriction of q_{ramp} to $[0, \pi]$, where it is the polynomial $s(t/\pi)$, enters Theorem 1; the plateau is part of the protocol's definition and enters only the statements on $[0, 2\pi]$ (Lemmas 3 and 5 and Proposition 1), not Theorem 1 or the numerical illustration, whose times all lie in $[0, \pi]$.

Definition 3 (centred and standardised force; degeneracy set). For a protocol q and a time t , $\dot{F}_q(t) = F_q(t) - \mathbb{E}F_q(t)$. The degeneracy set is $D_q = \{t : \text{Var } F_q(t) = 0\}$, and for $t \notin D_q$ the standardised force is $Z_q(t) = \dot{F}_q(t) / \text{sd } F_q(t)$.

Definition 4 (competitor classes). Every class below is a single model shared by *all* protocols in $\mathcal{X}$: its objects may depend on the prescribed path up to the current time, $q_{[0, t]}$, but not on a protocol label, and no Gaussian, Markov, finite-memory or stationarity restriction is imposed.

- $\mathcal{E}_1$ (additive exogenous): $F_q(t) = M_t[q] + \xi(t)$, with M an arbitrary deterministic causal functional and ξ one exogenous process whose law does not depend on q .

- $\mathcal{E}_2^\pm$ (shared signed-affine exogenous): $F_q(t) = M_t[q] + G_t[q]\xi(t)$, with M an arbitrary deterministic causal functional, G an arbitrary deterministic causal functional with $G_t[q] \in \mathbb{R} \setminus \{0\}$ (a *signed causal scale*: sign changes are allowed, zero is not), and ξ one exogenous process with one protocol-independent path law. Equivalently — under clamping, finite second moments and environment causality (the law of F_q on $[0, t]$ depends on q only through $q_{[0, t]}$), and at the level of finite-dimensional distributions — $\mathcal{E}_2^\pm$ membership holds exactly when (i) the degeneracy set $D_q = D$ is common to all q , and (ii) there is one shared deterministic causal sign functional $S_t : q_{[0, t]} \mapsto \{+1, -1\}$ such that, with $s_q(t) = S_t[q_{[0, t]}]$, the finite-dimensional laws of $s_q Z_q$ off D are the same for every $q \in \mathcal{X}$. *Prefix consistency* is part of membership: if $q_{[0, t]} = q'_{[0, t]}$ then $s_q(t) = s_{q'}(t)$.
- $\mathcal{E}_{\text{univ}}$ (universal causal exogenous): $F_q(t) = \mathfrak{F}_t[q, U]$, with U one exogenous random object of protocol-independent law and $\mathfrak{F}$ an arbitrary deterministic functional that is causal in q .

The harmonic class $\mathcal{H}$ (harmonic baths coupled linearly in the bath and system coordinates, with counterterm and a free-force law independent of the system's initial state) sits at the bottom, and the ladder $\mathcal{H} \subset \mathcal{E}_1 \subsetneq \mathcal{E}_2^\pm \subsetneq \mathcal{E}_{\text{univ}}$ holds, with both inclusions among the $\mathcal{E}$ -classes strict (Supplement S3). Every classical, deterministic, causal environment whose state obeys $\dot{Y} = B(Y, q)$ with force $C(Y, q)$ and initial state $\Psi(q(0), U)$ — with B jointly continuous and locally Lipschitz in Y , locally uniformly in q , C measurable, and no blow-up on $[0, T]$ — lies in $\mathcal{E}_{\text{univ}}$ (Proposition S3.7); model $\mathcal{D}$ is of this type (Lemma 3(i)). The question is therefore whether it lies in $\mathcal{E}_2^\pm$.

Definition 5 (response objects). Write $X_q^\varepsilon(t)$ for the position of one oscillator obeying (S1.2) from $z_0 = (a, b) \sim \rho$, and $x_0(t) = X_q^0(t)$ for the unforced trajectory. For the ramp protocol the first-order response $y_1 = \partial_\varepsilon X^\varepsilon|_{\varepsilon=0}$ solves

$$\ddot{y}_1 + (1 + 3x_0^2) y_1 = q_{\text{ramp}}(t), \quad y_1(0) = \dot{y}_1(0) = 0, \quad (\text{S1.7})$$

and we set

$$c(t) = \text{Cov}(x_0(t)^2, y_1(t)) = \mathbb{E}[(x_0(t)^2 - m_2) y_1(t)], \quad K(t) = 3c(t). \quad (\text{S1.8})$$

The second form of $c(t)$ uses $\mathbb{E} x_0(t)^2 = m_2$ for every t , which holds because the Gibbs law is invariant under the unforced flow.

Gibbs moments. Integration by parts against $e^{-a^2/2 - a^4/4}$ (the boundary terms vanish) gives

$$m_{k+1} + m_{k+3} = k m_{k-1}, \quad \text{in particular} \quad m_2 + m_4 = 1, \quad \text{Var}(x_0^2) = m_4 - m_2^2 = 1 - m_2 - m_2^2. \quad (\text{S1.9})$$

Odd moments vanish by symmetry, and the momentum b is standard normal and independent of a .

S1.3 Statements

Theorem 1 (finite-bath escape from the shared signed-affine class). Consider model $\mathcal{D}$ with bath size N_B , prepared in the Gibbs state at $\beta = 1$, together with the reference protocol $q_{\text{ref}} \equiv 0$ and the ramp protocol q_{ramp} , which equals $s(t/\pi)$ on $[0, \pi]$. There exists $\delta \in (0, 1]$ such that, for each fixed $t \in (0, \delta)$, there is a finite integer $N_0(t)$ with the following property: for every bath size $N_B \geq N_0(t)$, the interventional force laws of model $\mathcal{D}$ under the reference and ramp protocols admit no common representation in the class $\mathcal{E}_2^\pm$. Consequently, the interventional force family of model $\mathcal{D}$ over $\mathcal{X}$ lies outside $\mathcal{E}_2^\pm$ for all finite $N_B \geq N_0(t)$. The witness is the standardised skewness at the single time t : $\gamma_1[F_{\text{ref}}(t)] = 0$ for every N_B , whereas $\gamma_1[F_{\text{ramp}}(t)] < 0$ for every $N_B \geq N_0(t)$,

and both force variances are strictly positive. The numbers δ and $N_0(t)$ are existential: no value of either is claimed, and $N_0(t)$ is not claimed to be uniform in t .

Lemma 1 (exact cumulant identity). Under any clamp q and for every finite N_B , $F_q = \varepsilon \sum_j X_j$ with $X_1, \dots, X_{N_B}$ independent copies of the process X_q^ε . Consequently, for every order n and every finite tuple of times,

$$\kappa_n(F_q(t_1), \dots, F_q(t_n)) = N_B^{1-n/2} \kappa_n(X_q^\varepsilon(t_1), \dots, X_q^\varepsilon(t_n)). \quad (\text{S1.10})$$

Lemma 2 (reference parity). Under the reference protocol, $\kappa_3(F_{\text{ref}}(t)) = 0$ exactly for every t and every N_B , and $\text{Var } F_{\text{ref}}(t) = m_2 > 0$.

Lemma 3 (first-order third cumulant). Let $q = q_{\text{ramp}}$, so that $|q| \leq 1$ on $[0, 2\pi]$, and let $|\varepsilon| \leq 1$. Then (i) for each z_0 the clamped solution exists on $[0, 2\pi]$ and is C^∞ in (z_0, ε) ; (ii) y_1 solves (S1.7); (iii) every moment $\mathbb{E}[\prod_{i \leq 3} \partial_\varepsilon^{k_i} X^\varepsilon(t_i)]$ with $t_i \in [0, 2\pi]$ and $\sum_i k_i \leq 3$ is finite, uniformly in $|\varepsilon| \leq 1$, and differentiation in ε commutes with $\mathbb{E}$ up to third order; (iv) $\kappa_3(X^\varepsilon(t))$ is an odd C^3 function of ε with derivative $K(t)$ at $\varepsilon = 0$. Hence for each fixed $t \in [0, 2\pi]$ there is $\Theta(t) < \infty$ with

$$\kappa_3(F_{\text{ramp}}(t)) = \frac{K(t)}{N_B} + R_N(t), \quad |R_N(t)| \leq \Theta(t) N_B^{-2}. \quad (\text{S1.11})$$

Parts (i) and (iii), and their proofs, hold verbatim for any clamp with $|q| \leq Q$ on $[0, 2\pi]$, with constants depending on Q and on the bounds for the derivatives of q ; parts (ii) and (iv) are statements about the ramp protocol's response y_1 and its coefficient $K(t)$.

Lemma 4 (small-time sign). The Taylor coefficients of $c(t)$ satisfy $c_0 = c_1 = \dots = c_6 = 0$ and

$$c(t) = -\frac{\text{Var}(x_0^2)}{28\pi^3} t^7 + O(t^8), \quad \text{more precisely} \quad |c(t) - c_7 t^7| \leq \Lambda t^8 \quad \text{on } [0, 1], \quad c_7 = -\frac{\text{Var}(x_0^2)}{28\pi^3}. \quad (\text{S1.12})$$

Since $\text{Var}(x_0^2) > 0$, $c_7 < 0$, and there exists $\delta \in (0, 1]$ such that $c(t) < 0$, hence $K(t) = 3c(t) < 0$, for every $0 < t < \delta$.

Lemma 5 (positive variances). For every N_B and every $t \in [0, 2\pi]$, $\text{Var } F_{\text{ramp}}(t) = \text{Var } X^\varepsilon(t)$ is finite and strictly positive.

Lemma 6 (an invariant of $\mathcal{E}_2^\pm$). If a force family lies in $\mathcal{E}_2^\pm$ and $\mathbb{E} |F_q(t)|^3 < \infty$ for one protocol (hence, by the shared representation, for every protocol), then at every time t outside the common degeneracy set the absolute standardised skewness $|\gamma_1[F_q(t)]|$ takes the same value for every protocol q .

Proposition 1 (reservoir limit; $\mathcal{E}_1$ -type, finite-dimensional). (a) For each fixed $t \in (0, \delta)$, $|\gamma_1[F_{\text{ramp}}(t)]| = O(N_B^{-1})$ and tends to zero as $N_B \rightarrow \infty$. (b) For the reference and ramp protocols, and for any separately fixed admissible bounded clamp for which the moment estimates of Lemma 3 hold, and for each finite tuple of times in $[0, 2\pi]$, the centred force $\hat{F}_q$ converges in finite-dimensional distribution, as $N_B \rightarrow \infty$, to the centred Gaussian law with covariance

$$C_0(t_a, t_b) = \langle x_0(t_a) x_0(t_b) \rangle_{\text{Gibbs}}, \quad (\text{S1.13})$$

which is the same for every such protocol. No statement is made that one shared $\mathcal{E}_1$ representation holds uniformly over the whole infinite clamp class $\mathcal{X}$.

S1.4 Proofs

Proof of Lemma 1

Step 1 (independence). Under a clamp, every oscillator obeys the same deterministic equation (S1.2), driven by the same prescribed $\varepsilon q(t)$, from independent and identically distributed initial data. The solution map $z_0 \mapsto X_q^\varepsilon(t; z_0)$ is deterministic, so the trajectories are independent copies of one process, and $F_q = \varepsilon \sum_j X_j$ exactly. No approximation is involved.

Step 2 (cumulant generating function). For the vector $(F_q(t_1), \dots, F_q(t_n))$, independence gives $\log \mathbb{E} \exp(\sum_a \theta_a F_q(t_a)) = N_B \log \mathbb{E} \exp(\varepsilon \sum_a \theta_a X(t_a))$. Differentiating n times in θ at zero brings down ε^n , so $\kappa_n(F) = N_B \varepsilon^n \kappa_n(X) = N_B^{1-n/2} \kappa_n(X)$. $\square$

Why it matters. For $n = 2$ the force variance equals the single-oscillator variance, so the force fluctuation is of order one at every bath size; for $n = 3$ the force cumulant is $N_B^{-1/2}$ times the single-oscillator cumulant. Everything below is about one oscillator.

Proof of Lemma 2

The unforced vector field $(\dot{x}, \dot{p}) = (p, -x - x^3)$ is odd, so $(x, p) \mapsto (-x, -p)$ maps solutions to solutions, and ρ is even. Hence $x_0(t)$ has a law symmetric about zero at every t , $\mathbb{E} x_0(t) = 0$, and $\kappa_3(x_0(t)) = 0$. By Lemma 1, $\kappa_3(F_{\text{ref}}(t)) = N_B^{-1/2} \cdot 0 = 0$. The Gibbs law is invariant under the Hamiltonian flow, so $\text{Var } F_{\text{ref}}(t) = \text{Var } x_0(t) = m_2$, and $m_2 > 0$ because $a^2 \geq 0$ is not almost surely zero. $\square$

Proof of Lemma 3

Step 1 (energy bound and global existence). Along the forced flow $dH_0/dt = \varepsilon qp$, so $d\sqrt{H_0}/dt = \varepsilon qp/(2\sqrt{H_0}) \leq |\varepsilon|Q/\sqrt{2}$. Hence $\sqrt{H_0(t)} \leq \sqrt{E_0} + c_Q$ on $[0, 2\pi]$ with $c_Q = \sqrt{2}Q\pi$. Orbits stay in a compact set, so solutions are global on $[0, 2\pi]$, and because the vector field is polynomial in (z, ε) the flow is C^∞ in (z_0, ε) . This is (i).

Step 2 (variational equation). Differentiating (S1.2) in ε at $\varepsilon = 0$ gives (S1.7). The initial data do not depend on ε , because the preparation precedes the protocol and the coupling vanishes at $t = 0$; hence $y_1(0) = \dot{y}_1(0) = 0$. This is (ii).

Step 3 (moment bounds). Let $w_k = \partial_\varepsilon^k z$. Each w_k solves a linear equation $\dot{w}_k = A(t)w_k + f_k$ from zero, with $A = \begin{pmatrix} 0 & 1 \\ -1 - 3x^2 & 0 \end{pmatrix}$, $f_1 = (0, q)$, and, for $k = 2, 3$, inhomogeneities f_k that are polynomials in x and the lower-order w_j^x obtained by differentiating $-x^3$ (for example $f_2 = (0, -6x(w_1^x)^2)$). Since $x^4/4 \leq H_0$, $x^2 \leq 2\sqrt{H_0} \leq 2(\sqrt{E_0} + c_Q)$, so $\|A(t)\|$ and $|x|$ are bounded by a constant of the form $\text{const} \cdot (1 + \sqrt{E_0})$. Grönwall's inequality on $[0, 2\pi]$ then bounds every $|w_k|$, $k \leq 3$, and $|X^\varepsilon|$ by $P(E_0) e^{\lambda\sqrt{E_0}}$ for a polynomial P and a constant λ , uniformly in $|\varepsilon| \leq 1$.

Step 4 (integrability). Under ρ , $E_0 = H_0(z_0)$ and $e^{-E_0 + \lambda\sqrt{E_0}} \leq e^{\lambda^2/2} e^{-E_0/2}$; the phase-space area of $\{H_0 \leq e\}$ grows polynomially in e . So $\mathbb{E}[P(E_0)e^{\lambda\sqrt{E_0}}] < \infty$, and dominated convergence allows differentiating under $\mathbb{E}$ up to third order, with continuous derivatives. This is (iii).

Step 5 (oddness). The map $(x, p, \varepsilon) \mapsto (-x, -p, -\varepsilon)$ maps solutions to solutions and ρ is even, so $(X^{-\varepsilon}(t))_t$ has the law of $(-X^\varepsilon(t))_t$. Hence $\kappa_3(X^{-\varepsilon}) = -\kappa_3(X^\varepsilon)$: the third cumulant is odd in ε , vanishes at $\varepsilon = 0$, and $\mathbb{E} x_0 = 0$.

Step 6 (the derivative is K). Write $\kappa_3(X) = \mathbb{E}X^3 - 3\mathbb{E}X\mathbb{E}X^2 + 2(\mathbb{E}X)^3$ at one time t . Differentiate at $\varepsilon = 0$ using (iii) and $\partial_\varepsilon X|_0 = y_1$: the first term gives $3\mathbb{E}[x_0^2 y_1]$; the second gives $-3\mathbb{E}[y_1]\mathbb{E}[x_0^2]$, because the other term carries the factor $\mathbb{E}x_0 = 0$; the third vanishes for the same reason. So $\partial_\varepsilon \kappa_3(X^\varepsilon(t))|_{\varepsilon=0} = 3(\mathbb{E}[x_0^2 y_1] - \mathbb{E}[x_0^2]\mathbb{E}[y_1]) = 3c(t) = K(t)$. (For three distinct times the same computation gives the symmetric sum of three such covariances; Theorem 1 needs only the one-time case.)

Step 7 (Taylor in ε and conversion to N_B). Being C^3 and odd, $\kappa_3(X^\varepsilon(t)) = \varepsilon K(t) + R_3(\varepsilon)$ with $|R_3(\varepsilon)| \leq \Theta(t)|\varepsilon|^3$, where $\Theta(t) = \sup_{|\varepsilon| \leq 1} |\partial_\varepsilon^3 \kappa_3|/6 < \infty$ by (iii). Lemma 1 with $n = 3$ and $\varepsilon = N_B^{-1/2}$ gives $\kappa_3(F_{\text{ramp}}(t)) = N_B^{-1/2} \kappa_3(X^\varepsilon(t)) = K(t)/N_B + N_B^{-1/2} R_3(N_B^{-1/2})$, and the last term is bounded by $\Theta(t)N_B^{-2}$. $\square$

Remark. $\Theta(t)$ is finite but enormous (it comes from Grönwall); it certifies the *order* of the remainder, not its size at any particular N_B .

Proof of Lemma 4

Step 1 (pathwise smoothness). For fixed z_0 , x_0 solves a polynomial ODE and y_1 solves the linear ODE (S1.7) with polynomial forcing $s(t/\pi)$ on $[0, 1] \subset [0, \pi]$. Both are C^∞ , hence so is $g(t; z_0) = (x_0(t)^2 - m_2)y_1(t)$, and $c(t) = \mathbb{E}g(t; z_0)$.

Step 2 (a derivative bound on $[0, 1]$). Energy conservation gives $x_0^2 \leq 2\sqrt{E_0}$ and $p_0^2 \leq 2E_0$. With $k(t) = 1 + 3x_0^2 \leq \bar{k} = 1 + 6\sqrt{E_0}$ and $|q_{\text{ramp}}| \leq 1$, Grönwall gives $|y_1|, |\dot{y}_1| \leq e^{1+\bar{k}}$ on $[0, 1]$. Every time derivative of order at most eight of $x_0, \dot{x}_0, y_1, \dot{y}_1$ is, by repeated use of the equations, a polynomial in $(x_0, p_0, y_1, \dot{y}_1)$ and in derivatives of q_{ramp} , which are bounded on $[0, 1]$. So $\sup_{[0, 1]} |\partial_t^8 g(\cdot; z_0)| \leq D_8(z_0) = P(E_0)e^{1+\bar{k}}$ for a polynomial P .

Step 3 (integrability). Because $-E_0 + 6\sqrt{E_0} \leq -E_0/2 + \max_{u \geq 0} (6u - u^2/2)$ and phase-space volume grows polynomially in E_0 , D_8 is ρ -integrable; likewise $|\partial_t^k g(0; z_0)|$ is a polynomial in z_0 for $k \leq 7$.

Step 4 (Taylor first, expectation second). For each z_0 , $g(t) = \sum_{k=0}^7 g^{(k)}(0)t^k/k! + r(t; z_0)$ with $|r| \leq D_8(z_0)t^8/8!$. Take expectations of this pointwise identity; every term is integrable, so no interchange of derivative and expectation is needed:

$$c(t) = \sum_{k=0}^7 c_k t^k + R(t), \quad |R(t)| \leq \Lambda t^8, \quad c_k = \frac{\mathbb{E}[g^{(k)}(0)]}{k!}, \quad \Lambda = \frac{\mathbb{E}D_8}{8!} < \infty. \quad (\text{S1.14})$$

Step 5 (the coefficients). Expand $x_0(t) = \sum_k \alpha_k t^k$ and $y_1(t) = \sum_k \eta_k t^k$. The equations give the recursions

$$(k+2)(k+1)\alpha_{k+2} = -\alpha_k - (x_0^3)_k, \quad (k+2)(k+1)\eta_{k+2} = -\eta_k - 3(x_0^2 y_1)_k + q_k, \quad (\text{S1.15})$$

with $\alpha_0 = a$, $\alpha_1 = b$, $\eta_0 = \eta_1 = 0$, where $(\cdot)_k$ denotes the coefficient of t^k and q_k those of $s(t/\pi)$: $q_3 = \frac{10}{\pi^3}$, $q_4 = -\frac{15}{\pi^4}$, $q_5 = \frac{6}{\pi^5}$, all others zero through the orders needed. Because $\eta_j = 0$ for $j \leq 4$, the first non-zero response coefficients are forced by q alone until the $x_0^2 y_1$ term switches on:

$$\eta_5 = \frac{1}{2\pi^3}, \quad \eta_6 = -\frac{1}{2\pi^4}, \quad \eta_7 = -\frac{1+3a^2}{84\pi^3} + \frac{1}{7\pi^5}. \quad (\text{S1.16})$$

Writing $x_0(t)^2 = \sum_i d_i t^i$, one has $d_0 = a^2$, $d_1 = 2ab$ and, using $\alpha_2 = -\frac{a^3}{2} - \frac{a}{2}$, $d_2 = -a^4 - a^2 + b^2$. The coefficient of t^n in $c(t) = \mathbb{E}[x_0^2 y_1] - m_2 \mathbb{E}[y_1]$ is

$$c_n = \text{Cov}(a^2, \eta_n) + \sum_{i=1}^n \mathbb{E}[d_i \eta_{n-i}]. \quad (\text{S1.17})$$

- For $n \leq 4$ every η_j that appears vanishes, so $c_0 = \dots = c_4 = 0$.
- $c_5 = \text{Cov}(a^2, \eta_5) = 0$ and $c_6 = \text{Cov}(a^2, \eta_6) + \eta_5 \mathbb{E}[2ab] = 0$, because η_5, η_6 are constants and $\mathbb{E}[ab] = \mathbb{E}a \mathbb{E}b = 0$.
- For $n = 7$ only $i \leq 2$ can contribute. The $i = 1$ term is $\eta_6 \mathbb{E}[2ab] = 0$. The $i = 2$ term is $\eta_5 \mathbb{E}[b^2 - a^2 - a^4] = \eta_5(1 - m_2 - m_4) = 0$ by (S1.9). What remains is $\text{Cov}(a^2, \eta_7)$; only the a^2 part of η_7 (coefficient $-\frac{1}{28\pi^3}$) is random, so

$$c_7 = \left(-\frac{1}{28\pi^3}\right) \text{Var}(a^2) = -\frac{\text{Var}(x_0^2)}{28\pi^3} = \frac{m_2^2 + m_2 - 1}{28\pi^3}. \quad (\text{S1.18})$$

The constant parts of η_7 , including the π^{-5} term, cancel between $\mathbb{E}[a^2 \eta_7]$ and $m_2 \mathbb{E}[\eta_7]$; this cancellation is a useful arithmetic check. The same coefficients, through t^{14} , are produced by an exact computer-algebra recursion (Supplement S2).

Step 6 (sign). a has a strictly positive density, so a^2 is not almost surely constant and $\text{Var}(x_0^2) = \text{Var}(a^2) > 0$; hence $c_7 < 0$. No numerical value of m_2 is needed.

Step 7 (the window). Put $\delta = \min(1, |c_7|/(2\Lambda))$. For $0 < t < \delta$, (S1.14) gives $c(t) \leq c_7 t^7 + \Lambda t^8 = t^7(c_7 + \Lambda t) < t^7 c_7 / 2 < 0$. $\square$

Remarks. δ is existential, because Λ is an existence bound that is not computed. No complex-time radius and no all-orders expansion is used.

Proof of Lemma 5

By Lemma 1 with $n = 2$, $\text{Var} F_{\text{ramp}}(t) = \text{Var} X^\varepsilon(t)$, which is finite by Lemma 3(iii). The time- t map $z_0 \mapsto z(t; z_0, \varepsilon)$ of the smooth global flow is a C^1 diffeomorphism of $\mathbb{R}^2$, and ρ has a density. A diffeomorphism carries a law with a density to a law with a density, so $(x, p)(t)$ has a density, $x(t)$ is not almost surely constant, and the variance is strictly positive. $\square$

Proof of Lemma 6

Suppose $F_q(t) = M_t[q] + G_t[q] \xi(t)$ with one shared law for ξ . For a clamped protocol, $M_t[q]$ and $G_t[q]$ are deterministic numbers, and $G_t[q] \neq 0$. Then $\text{Var} F_q(t) = G_t[q]^2 \text{Var} \xi(t)$, so the degeneracy set is that of ξ for every q , and off it

$$\kappa_3(F_q(t)) = G_t[q]^3 \kappa_3(\xi(t)), \quad \gamma_1[F_q(t)] = \text{sgn}(G_t[q]) \gamma_1[\xi(t)]. \quad (\text{S1.19})$$

Hence $|\gamma_1[F_q(t)]| = |\gamma_1[\xi(t)]|$ for every q . $\square$

Why the absolute value. The sign of G is allowed to depend on the protocol (that is what “signed” means), and a sign flip reverses the skewness. The absolute standardised skewness is invariant under every deterministic location shift, rescaling and sign flip, so it is exactly the kind of quantity that a shared signed-affine model must hold fixed across protocols. This is the one-time case of the reflection-orbit criterion for $\mathcal{E}_2^\pm$.

Proof of Theorem 1

Take δ from Lemma 4 and fix $t \in (0, \delta)$.

1. *Non-zero first-order coefficient.* By Lemma 4, $K(t) = 3c(t) < 0$.
2. *Threshold.* With $\Theta(t)$ from Lemma 3, set $N_0(t) = \lfloor \Theta(t)/|K(t)| \rfloor + 1$, a finite integer. For every $N_B \geq N_0(t)$ we have $N_B > \Theta(t)/|K(t)|$, so $|R_N(t)| \leq \Theta(t)N_B^{-2} < |K(t)|N_B^{-1}$, and (S1.11) gives $\kappa_3(F_{\text{ramp}}(t)) \neq 0$ with the sign of $K(t)$, that is, negative.
3. *Standardisation is legitimate.* By Lemma 5, $0 < \text{Var } F_{\text{ramp}}(t) < \infty$, so $\gamma_1[F_{\text{ramp}}(t)] < 0$. By Lemma 2, $\text{Var } F_{\text{ref}}(t) = m_2 > 0$ and $\gamma_1[F_{\text{ref}}(t)] = 0$. In particular t is a non-degenerate time for both protocols, so the conclusion below is not a degeneracy artefact.
4. *Contradiction.* If the two force laws had a common $\mathcal{E}_2^\pm$ representation, Lemma 6 would give $|\gamma_1[F_{\text{ramp}}(t)]| = |\gamma_1[F_{\text{ref}}(t)]| = 0$, contradicting step 3.

Hence no common $\mathcal{E}_2^\pm$ representation exists for $N_B \geq N_0(t)$; a fortiori the whole interventional family over $\mathcal{X}$ lies outside $\mathcal{E}_2^\pm$. The witness is a violation at a single non-degenerate time: the standardised law of the ramp protocol lies outside every reflection image of the reference law. $\square$

Quantifiers, made explicit. $\exists \delta > 0$ such that $\forall$ fixed $t \in (0, \delta)$, $\exists N_0(t) < \infty$ such that $\forall N_B \geq N_0(t)$: $\kappa_3(F_{\text{ramp}}(t)) \neq 0$. The threshold depends on t because $\Theta(t)$ does; no uniform N_0 over the interval is claimed. Nothing in the proof uses a numerically chosen time.

Proof of Proposition 1

The proof, which uses Lemmas 1–3, the window δ of Lemma 4 for the domain of part (a), and a standard multivariate central limit theorem, is given in Supplement S4. Part (a) reads, explicitly,

$$\gamma_1[F_{\text{ramp}}(t)] = \frac{K(t)}{m_2^{3/2} N_B} + o(N_B^{-1}) \rightarrow 0. \quad (\text{S1.20})$$

S1.5 What is not claimed

- No numerical value of δ or of $N_0(t)$; no uniform threshold in t .
- No statement that any particular numerical time, including $t_\star = 0.5$ used in Section 6, lies inside $(0, \delta)$.
- No topological or closure statement about membership in $\mathcal{E}_2^\pm$, and no claim that the finite- N_B effect stays observable as N_B grows.
- No escape from $\mathcal{E}_{\text{univ}}$: model $\mathcal{D}$ is representable there.
- No shared $\mathcal{E}_1$ representation uniformly over the full clamp class $\mathcal{X}$.